INCH POUND

A-A-59210
11 August 2003

COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, MB-4

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the requirements for one type of MB-4 aircraft towing tractor, hereafter referred to as tow tractor. The tow tractor must be capable of towing trailers up to 80,000 lbs, and military aircraft up to 175,000 pounds on dry level concrete.

2. SALIENT CHARACTERISTICS. Failure of the tow tractor to meet the salient characteristics of this CID shall be cause for rejection of the tow tractor. The tow tractor shall be free from faulty construction and any defect that affects appearance, serviceability, and durability. The tow tractor shall be representative of the quality and workmanship found in tow tractors offered for new commercial sale.

a. Tow tractors shall have a minimum gross vehicle weight (GVW) of 19,000 pounds.
b. Tow tractors shall have a minimum drawbar pull (DBP) of 16,000 pounds on dry level concrete.
c. Tow tractor shall have a diesel engine and power train sufficient to develop the required drawbar pull. The vehicle shall operate satisfactorily in high temperatures up to 125 degrees F. Diesel engine shall be unaffected by the use of Jet A-1 (JP-8) fuel. Power train shall feature a standard winterization package that is good for starting down to minus 20F, or when specified (see 6.2 c), a heavy-duty winterization package good for storage down to minus 65F and starting to minus 40F. When specified (see 6.2 d), tow tractor shall comply with California regulations governing air pollution control in effect on the date of manufacture. Engine shall have an automatic engine shutdown system. Transmission shall be of the power-shift type, with an integral no-spin, inter-axle differential. Axles shall be planetary-drive/steer axles.
d. Tow tractor shall have four-wheel drive and four-wheel steering. Four-wheel steering shall feature both crab and coordinated steering. Tow tractors which feature electronically controlled four wheel steering, or electronically controlled diesel engines

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: 295 Byron Street, WR-ALC/LEE Robins AFB, GA 31098-1611

AMSC N/A FSC 1740

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and transmissions, shall comply with EMI requirements of MIL STD 461E for Air Force ground equipment. For electronically controlled diesel engines and transmissions, the tow tractor shall be tested for Radiated Emission per RE102 of MIL STD 461E and Radiated Susceptibility per RS103 of MIL STD 461E. For electronically controlled steering, the tow tractor shall be tested for Conducted Susceptibilty per CS114, CS115 and CS116 of MIL STD 461E. Wheel-wells shall be designed so that tire chains may be used without interference.

e. Tow tractor shall be capable of traveling a minimum of 25 MPH forward.

f. Tow tractor shall feature four-wheel hydraulic operated brakes with power assist.

g. When specified (see 6.2 e), the tow tractor shall have an air system for towing the Air Force's MHU-196 and MHU-204 trailers with air brakes. The air system shall be in accordance with applicable commercial standards for over the road truck tractors. The glad hands shall be in accordance with SAE J318. The tow tractor's air system shall supply air to the trailer with sufficient speed and volume so as to work the tow tractor and trailer brakes simultaneously. The tow tractor/trailer combination shall stop straight in a 12 foot wide corridor without evidence of jackknifing. The MHU-196/204 trailers have 1/4" ID service brake air lines and 3/8" ID parking brake air lines. The trailers have four wheel brakes. The distance from the rear of the tow tractor to first set of wheels on the trailer is approximately 10 feet, and from the first set of wheels to the second set of wheels is approximately 30 feet. There are four air brake chambers, each with a volume of 28.8 cubic inches for the service brakes, and 45.0 cubic inches for the parking brakes. The braking system working pressure is 90 psi. The MHU-196 trailer is towed at 10 mph loaded, and 20 mph unloaded. The MHU-204 trailer is towed at 10 mph loaded or unloaded. The maximum weight of the loaded MHU-196/204 trailer is 80,000 lbs. The government shall test the air equipped tow tractor with the above mentioned trailers and work with the contractor to make any necessary design adjustments to the air/brake system to ensure acceptable braking performance.

h. Tow tractor shall be equipped with a mechanically actuated disc type parking brake. The parking brake system shall include an audible chime and visual warning light that engages when the parking brake is on and the vehicle is placed in forward or reverse gear.

i. Tow tractor shall feature a 12V electrical system.

j. Tow tractor shall have an all welded, one piece frame structure.

k. Tow tractor shall be equipped with an enclosed two-man cab. Interior cab sound level shall not exceed 84 dB (A) with doors and windows closed. The first entrance step to the cab shall not exceed 18 inches in height from the ground. Cab instrumentation shall be standard commercial installation to include an hour meter, odometer, tachometer and speedometer.

l. Tow tractor cab shall be equipped with wipers, mirrors, visors, heater/defroster, ventilating fan, interior light, seat belts and other equipment as found in standard commercial offerings. The cab shall feature a rear sliding window and a 12V power outlet (cigarette lighter type).

m. A flashing beacon light shall be mounted on the cab roof.

n. Tow tractor shall be equipped with front and rear pintle hooks that incorporate a cotter pin as a secondary-locking device, and conforming to CID A-A-52550 Type 1, Class 2

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(old MS51335-2). The pintles shall be located on the centerline of the tractor and 14 inches above the ground. Additional mounting holes for a pintle at 25 inches above the ground shall be provided (front and rear). For the front pintle positions, the pintle assembly and mounting structure shall be designed to accommodate the greatest force exerted by the drawbar with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load and unload operations. For the rear pintle positions, the pintle assembly and mounting structure shall comply with the following:

1. The rated load shall be the load combination of forces that the pintle assembly and structure must support or resist in a static state.
2. The dynamic load shall be the rated load with adjustments for loads and accelerations, in all directions, encountered during air/ground transport and towing operations, and the shock loads associated with mate, demate, load, and unload operations.
3. The design load shall be based on the rated load multiplied by a factor of three, or on the dynamic load multiplied by a factor of two, which ever is greater. This design load will be considered the minimum load for attaining the design stress levels.

o. When an air system for towing trailers is specified (see 6.2 e), tow tractor shall be equipped with a rear trailer emergency breakaway cable hookup, capable of mating with NAS 1281C10 spring snap hook. The breakaway chain attachment shall be installed on the bumper or frame within 9 inches of the pintle assembly. It shall not be installed on the pintle hook or pintle hook assemblies.

p. When an air system for towing trailers is specified (see 6.2 e), tow tractor shall have two rear safety chain attachment points, one located on each side of the pintle hook, symmetrical about the pintle hook and shall be located 14 inches high for use during towing with rear pintle only. The attachment points shall be located a distance of not greater than 12 inches from the pintle hook and shall in no way be a part of, or attached to, the pintle hook. Each of the attachment points shall be capable of withstanding a shock load of 16,000 pounds and shall have the ability for standard safety chain hooks to be attached. The inside radius of the safety chain hook is between one inch and three inches.

q. Tow tractor lighting shall be equivalent to the standard commercial offering to include a rear coupling light, parking lights, head lights (high and low beam), brake lights, turn signals, and hazard lights.

r. Tow tractor shall be equipped with two 5 lb dry chemical type fire extinguishers with mounting brackets.

s. As specified (see 6.2 f), tow tractor shall be painted Dark Green (Color 24052, FED-STD-595) or Desert Sand (Color 23448, Fed-STD-595) using offeror's standard commercial paint system.

t. Tow tractor shall be rust proofed and undercoated in accordance with FED-STD-297. The level of corrosion treatment shall be tropical, and the use of proprietary material shall require Government approval.

u. Tow tractor shall be so constructed as to provide maximum visibility in all directions. Visibility of the front and rear pintle hooks shall be provided either directly or via mirrors from the driver’s position. If required, one mirror per pintle hook shall be provided to complete this visibility requirement.

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v. Tow tractors offered under this specification shall comply with applicable safety requirements imposed on vehicles of this type in the commercial market.
w. Tow tractor shall be air transportable on C-130, C-141, C-5 and C-17 aircraft in accordance with MIL-HDBK-1791.
x. The tractor shall have an identification plate and a rear license plate mounting bracket. The identification plate shall include as a minimum the following.
   1. Vehicle Nomenclature
   2. Manufacturer's Model Number
   3. Vehicle Identification (serial) Number or VIN
   4. Manufacturer's address and phone number
   5. Date of Manufacture
   6. National Stock Number
   7. Contract Number
   8. Registration Number

y. The vehicle shall be provided with a lubrication data plate that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

z. A transportation data plate shall be provided. The plate shall contain at least the following information.
   1. Side and rear silhouette views of the vehicle
   2. Horizontal and vertical location of the center of gravity of the vehicle in air transportable configuration, marked on the silhouette views
   3. Shipping weight
   4. Loading cubage
   5. Overall height, width, and length
   6. Front and rear axle loads
   7. Tie down information

3. RECLAIMED MATERIAL. The contractor shall utilize reclaimed materials to the maximum extent practical. Used, rebuilt, or remanufactured components shall not be incorporated in the vehicle.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be substantially the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Commercial item requirement. The vehicle furnished must meet the "commercial item" definition, as specified in FAR 2.101, as of the date of award. The offeror/contractor shall provide two copies of their commercial, descriptive catalogs with their offer, as specified in Clause I106. The offeror/contractor shall identify all modifications made to their commercial

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model in order to meet the performance and descriptive requirements of the CID or the referenced documents. In regard to the offered item, the offeror/contractor shall identify any enhancements or improvements to the performance requirements of paragraph 2 of the CID or the referenced documents.

4.3 Contractor requirements. The contractor shall provide copies of the truck manuals for review 30 days prior to the time of tests or demonstration of the first truck. The contractor's parts manuals shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. The contractor shall also provide with each truck a product familiarization videotape that verbally and visually provides all information for the operation and routine maintenance of the truck and its components, using the commercial manuals as a baseline.

4.4 Verification. Prior to delivery of the first truck, the contractor shall demonstrate or test, at his facility, each requirement of the CID or referenced documents, prior to delivery.

4.5 Reconfiguration Criteria. Failure to comply with the requirements of this CID, reference documents, or the tests shall be cause for reconfiguration and re-demonstration. Causes for reconfiguration shall include: structural failure; non-correctable misalignments; component interference; conditions presenting a safety hazard to the user or maintainer: instability during operation, spillage of fuel or coolant, and overheating.

4.1 User Tests. If specified in the contract, contractor will provide a specified number of sample units to be placed in service at representative Air Force bases. These sample units will be used interchangeably with existing MB-4s to determine their fitness for use. The period of testing and evaluation will be 90 days.

5. PACKAGING, PACKING, and MARKING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order (see 6.2b).

6. NOTES.

6.1 Addresses for obtaining copies of referenced documents.

Copies of Department of Defense specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

Application for copies of Society of Automotive Engineers (SAE) documents should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendael, PA 15096-0001.

6.2 Ordering data. The purchaser shall specify the following at the time of purchase:

a. Title, number and revision letter, and date of this document.

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b. Packaging requirements (see 5.1).
c. If heavy-duty winterization is required (see 2c).
d. When the tow tractor shall comply with California regulations governing air pollution control in effect on the date of manufacture (see 2c).
e. When an air system for trailer towing is required (see 2g, 2o, & 2p).
f. Color required (see 2s).

6.3 Subject (key word) listing.

Aircraft parking
Ground handling
Tug

Custodians:
Air Force – 99

Preparing activity:
Air Force – 84

Agent
Air Force – 99

Civilian coordinating activity:
GSA/FSS

(Project 1740-0189)

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